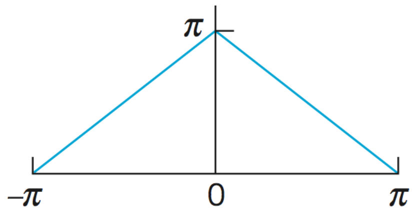
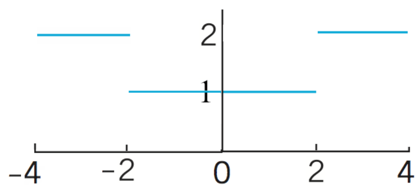
Answer the following and show the integration details! Don't just plug it into a CAS.

Problem 1: Determine the Fourier Series for each of the periodic functions depicted in the graphs below. Assume each of them periodically extend along the x-axis.

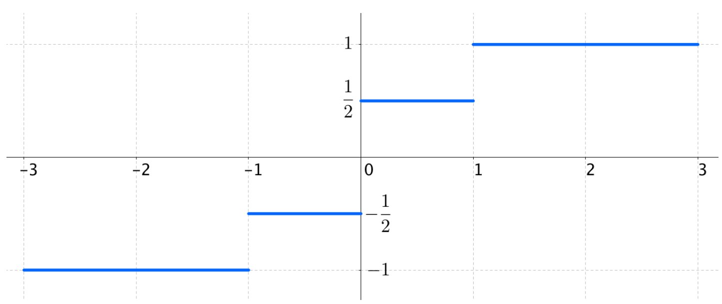
(i)



(ii)



(iii)



Problem 2: Let $f(x) = x^2$ on $-1 \leq x \leq 1$ and define $f(x)$ to have period 2 ($f(x) = f(x+2)$). Show that the Fourier Series for $f(x)$ is given by:

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(n\pi x)$$

Extra Credit: Use the result of **problem 2** to derive the summation identities

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6} \qquad \text{and} \qquad \sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$